Class-46 SQL 26/06/2025

DATA BASE

* Database runs on a server.
* DBMS - Data Base Management System.
  → It is used to manage a database (where we store data) b/w user & database.
  → File System DBMS:
* Redundancy - There is no change in data in another files when we update in one file.
  ↓ to overcome this problem.
* RDBMS: Relational DataBase Management System.
  → Network system.

27/06/2025

Class:- 47

RDBMS

* SQL - Structured Query Language (Not a programming language) (script) → Not a database
* It stores data in the form of tables (have rows & Columns).
* We can create, read, update, delete the data in tables.
* These operations are called CRUD Operations.
* NOSQL: Not Only Structured Query Language. → Not a database
* It stores data in key-Value pair (like HashMap) - JSON format ↓ (Java Script Object Notation)

```
{
  key1 : value1,
  key2 : value2
}
```

* It has no tables. f ~~faster~~
* It is faster than tables (due to data check with keys)
* It has MongoDB, Cassandra etc databases support NOSQL.
* MYSQL - It is a database.

MYSQL — Case insensitive.

1) DDL - Data Definition Language.
2) DML - Data Manipulation Language.
3) DQL - Data Query language
4) DCL - Data Control language
5) TCL - Transaction Control language.

* DDL: It will define actual structure of table.

It has, CREATE → operations * It modifies the structure
ALTER(Modify) but not table.
DROP
TRUNCATE
RENAME

* DML: When we playing with actual data it is used.

It has, INSERT * We can change only
UPDATE → operations data inside table, we
DELETE can't change table.

* DQL: It is used to read the data from table.

It has, SELECT

* DCL: We can control the data to access (by us(or) others).

It has, GRANT → Commands
REVOKE

* TCL: * When we turn on autocommit command, we can't roll back the data.

It has, COMMIT (save)
ROLLBACK (similar to UNDO operation)

Class-48 30/06/2025

DDL:

To work with SQL:

* Create a schema (project)/database.
* We can create multiple tables inside schema.
* To create a schema,
* SQL is case insesitive.

Syntax:

$\boxed{\text{Create database database\_name;}}$

* We use lower case. We differentiate the words in

SQL by using (_) underscore.

Ex: first_name.

Ex: * We can't create database with same name.

$\boxed{\text{Create}}$ database flm-adv-java;

$\boxed{\text{Show}}$ databases;

// To run. in mysQL workbench, — click on thunder symbol.

// To verify - we can see in schemas.

Command Prompt:

mysQL > show database; → to know what schemas we have in workbench.

*// To remove database, we use command DROP.

Syntax:

drop database flm_adv_java;

* ~~To create tables~~ in schema.

* Datatypes in SQL:

1. int / INTEGER ⇒ smallint, bigint
2. byte / BYTE ⇒ tiny int (-128 to 127 range).
3. decimal (we can define precession to decimal)
   decimal(5,2) ⇒ 100.25 (after . it has only two numbers). & one number but must has 5 numbers)
4. Numeric
5. Float
6. Double
7. Varchar (use to store multiple characters) (ASCII) - we can give size to it. It will accept characters according to that size, if we mention size.

* NVarchar
✓ (Advanced)
National Varchar

8. Text (use to store long text).
   → tinytext
   → mediumtext
   → longtext

* NVarchar - Supports many languages. (unicode)